Statement of Novelty

Fixed Points and Certified Boundedness of Unrolled Born/Contrast-Source Inverse Scattering Networks

Longgang Wang, Guoqiang Wang, and Nianfeng Shi

Statement of Novelty. The novelty lies in an obstruction rather than in a performance claim. For the Born/contrast-source family of unrolled inverse-scattering networks we show that the Jacobian block coupling contrast perturbations into the contrast-source channel equals a scaled diagonal of the total field, so a closed-form lower bound on the layer's largest singular value follows at once. That bound implies a step-size threshold above which no correction confined to the contrast channel can make the layer contractive, which rules out the per-layer Euclidean condition used elsewhere in the unrolling literature. Certification is therefore rebuilt layer by layer: a field-subsystem contraction condition, an invariant compact set under a nonexpansive projection, and an exact finite-depth noise account that keeps the non-normal factor a geometric series omits. The impact for readers is a reusable template that separates what can be certified from what is only measured, and that identifies the projection, not the learned correction, as the ingredient that bounds the rollout.